

Question number	Answer	Additional guidance	Mark
8(a)	An answer that includes three of the following points, with at least one similarity and one difference: Similarities: • both contain carbon and hydrogen (and oxygen) (1) • both contain covalent bonds (only) (1) Differences: • unsaturated fatty acids contain carbon-carbon double bonds but saturated fatty acids have only single carbon-carbon double bonds (1)	DO NOT PIECE TOGETHER if in two separate descriptions. ACCEPT hydrocarbon chain / COOH group DO NOT ACCEPT ester bonds ACCEPT saturated fatty acids do not contain carbon-carbon double bond C = C and C - C DO NOT ACCEPT saturated fatty acids have no double bonds	(3)

	- unsaturated fatty acids have {fewer hydrogens (for same number of carbons) / lower hydrogen to carbon ratio / higher carbon to hydrogen ratio} (1) - unsaturated fatty acids are {kinked / bent} and saturated fatty acids are {straight / linear} (1)	ACCEPT converse for saturated fatty acids	
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Question number	Answer	Additional guidance	Mark
8(b)(i)	The correct answer is A <i>B is incorrect because $41701 \div 68786 = 0.606242549355973$, so ratio is 0.6 : 1</i> <i>C is incorrect because $41701 \div 68786 = 0.606242549355973$, so ratio is 0.6 : 1</i> <i>D is incorrect because $41701 \div 68786 = 0.606242549355973$, so ratio is 0.6 : 1</i>		(1)

Question number	Answer
*8(b)(ii)	Animal products: • how much animal products / fats / cholesterol / LDL / HDL do you consume in a week? • because cholesterol increase risk of atheroma • high HDL : LDL reduces risk of CVD • has your consumption of fatty acids decreased? • because this could reduce the risk without any avocados being eaten Other dietary factors: • how much {fibre / vegetables / fruit / cereal} do you consume each week? • because fibre reduces cholesterol levels • how much food containing antioxidants do you consume? • antioxidants reduce damage to endothelial cells • how much salt do you consume? • high salt intake is linked to high blood pressure • how much alcohol do you consume in a week? • alcohol affects liver function / increase blood pressure / affects the LDL:HDL ratio • is your calorie consumption greater than your energy usage? • obesity is a risk factor for CVD Avocados: • do you eat avocados every week? • did you have adverse reactions to the avocado? • because study needed to ensure that they were being eaten on a weekly basis • have you replaced animal products with avocados?

	• because part of the benefit of eating avocados is to reduce saturated fatty acids in diet • the effects of the diet need to be compared to previously • how many avocados do you eat each week? • to see if the number of avocados affect the results Non-dietary questions: • do you smoke / how many cigarettes do you smoke / do you have high blood pressure? • because smoking increases blood pressure / damages endothelial lining • do you have diabetes / family history / BMI / age? • these increase the likelihood of CVD
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			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	Demonstrates isolated elements of biological knowledge and understanding to the given context with generalised comments made. Vague statements related to consequences are made with limited linkage to a range of scientific ideas, processes, techniques and procedures. The discussion will contain basic information with some attempt made to link knowledge and understanding to the given context.	Questions suggested, relating to non-diet CVD risk factors with some justification 1 mark = 1 question suggested 2 marks = 1 question suggested which does not relate to diet with justification
Level 2	3-4	Demonstrates adequate knowledge and understanding by selecting and applying some relevant biological facts / concepts. Consequences are discussed which are occasionally supported through linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows some linkages and lines of scientific reasoning with some structure.	Questions suggested that relate to diet with justification 3 marks = 1 question which relate to diet justified 4 marks = 2 questions which relate to diet justified
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting and applying relevant biological facts / concepts. Consequences are discussed which supported throughout by sustained linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows a well-developed and sustained line of scientific reasoning which is clear and logically structured.	Questions appropriate to this investigation with justifications 5 marks = 3 questions which relate to diet with justification justifications 6 marks = 4 questions which relate to diet with justifications

Question number	Answer	Additional guidance	Mark
8(b)(iii)	An answer that includes at least one reason for it being valid and one reason for it not being valid for full marks to be awarded: Valid because : • large sample size (1) • represents both {men and women / a range of ages} (1) • ran for {a long period of time / 30 years} (1) • participants had no signs of CVD or stroke (1) Invalid because : • no control group (1)	NB must state if the reason makes it valid or not ACCEPT converse idea for invalid IGNORE number of men and women were different ACCEPT longitudinal study IGNORE healthy	(4)

	• no indication that {lifestyles are similar / there were no (other) underlying diseases} (1) • {diets / number of avocados} not controlled (1) • responses to questionnaire {are subjective / may not be truthful} (1)	ACCEPT named lifestyle lifestyles not controlled other factors that increase risk of CVD difficult to control ACCEPT frequency of questionnaire is low / questionnaire is every 4 years	
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